

Linear regression assumptions

Every model made by some assumptions, if that assumptions are not validate

The model performance become less

Linear regression assumptions is divided into two ways

- *Based on features or input columns*
- *Based on Errors or Residuals*

Assumptions based input columns:

- ***Linearity:***
 - *There is a linear relationship between dependent(y) and independent variable(x)*
 - *Model name is Linear regression*
 - *it is an important property to maintain linear relationship*

How to check:

- *Scatter plots are the best to identify the linear relationship*
- *we cant draw more than 3D data*
- *Scatter plots between individual features vs output*

We want Linearity

if Linearity property not holds model performace become poor

- *then implement polynomial regression*
 - *apply transformations*
 - *try with another regression methods(DT regression, KNN Regressor etc)*
-
- ***Multicollinearity:***
 - *Input variables should be independent each other*
 - *If input columns having relation then this multicollineraity*
 - *Multicollineraity is similar to correlation*
 - *Any model: It is better to have input columns No Multicollinearity*

We don't want Multicollinearity

- *How to check*

Math analogy

x_1	x_2	$y = b_0 + b_1 * x_1 + b_2 * x_2$
2	4	
4	8	
6	12	
8	16	

x_1 and x_2 both are dependent : $x_2 = 2 * x_1$

$$y = b_0 + b_1 * x_1 + b_2 * 2x_1$$

$$y = b_0 + x_1[b_1 + 2b_2]$$

$$b_1 + 2b_2 = 0$$

maths says, if you want find two unknowns we need two equations

but above we have one equation and two unknowns : Maths fail

VIF: Variance inflation factor

VIF will give which variable cause the multicollinearity

$$VIF = \frac{1}{1 - R_{square}}$$

Procedure:

we already know what is R_{square}

R_{square} : *Explanability*

How much input variable explaining about variance in output variable

Assume that we have 3 input columns are there: x_1, x_2, x_3

$$Y = b_0 + b_1 * x_1 + b_2 * x_2 + b_3 * x_3$$

Main goal : Identify the Multicollinearity between input columns

For that we need to calculate VIF value

VIF value depends on R_{square} value

Step – 1: Consider x_1 as output variable ,

x_2 and x_3 consider as input

$$x_1 = b_0 + b_2 * x_2 + b_3 * x_3$$

the above equation will give R_{square}

R_{square} will give VIF value

$$\text{assume: } R_{square} = 0.9 \quad VIF = \frac{1}{1-0.9} = \frac{1}{0.1} = 10$$

Step – 2: Consider x_2 as output variable ,

x_1 and x_3 consider as input

$$x_2 = b_0 + b_1 * x_1 + b_3 * x_3$$

the above equation will give R_{square}

R_{square} will give VIF value

$$R_{square} = 0.95 \quad VIF = \frac{1}{1-0.95} = \frac{1}{0.05} = 20$$

Step – 3: Consider x_3 as output variable ,

x_1 and x_2 consider as input

$$x_3 = b_0 + b_1 * x_1 + b_2 * x_2$$

the above equation will give R_{square}

$$R_{square} = 0.80 \quad VIF = \frac{1}{1-0.8} = \frac{1}{0.2} = 5$$

Conclusion:

Depends on problem statement, will have the VIF cutoff

*In industry $VIF > 10$ we can consider as Multicollinearity is there
 x_2 gives $VIF = 20$, so x_2 has more multicollinearity with x_1 and x_3
we will not consider x_2 for model development*

We might do this

- 1) Remove the variables*
- 2) PCA: Principle component analysis will make columns as independent*
- 3) Ridge regression also use unimportant columns but it will less weightage*

So Multicollinearity concept (VIF) also used as Feature selection

Till now the Feature selection methods are

- 1) p - value*
- 2) Adj R _Square*
- 3) VIF*

Assumptions based on Errors or Residuals:

- *NO Autocorrelation*
Means errors are not correlated each other
- *Homoscedasticity:*
*The variance of errors is same across all independent variables
which means error variance should be constant*
- *Errors Follows Normality*
The distribution of errors follows Normal distribution

NO Autocorrelation

Imagin that we have a data of 10 observations

each 10 observations we have 10 actual output

10 observations we pass to model , then model will give 10 prediction

we will get 10 errors

	X	y_a	y _p	Σ
1		$\checkmark$	$\checkmark$	e_1
2				e_2
3				e_3
4				
10				e_n

Auto correlation:

Finding the relation between the values of same column

Errors should not have autocorrelation

How to check:

Durbon watson test(DW test) : Statistical test

$e_1, e_2, e_3, e_4, \dots \dots e_n$

Step – 1: Difference between the errors

$$d_1 = e_2 - e_1$$

$$d_2 = e_3 - e_2$$

$$d_3 = e_4 - e_3$$

$$d_t = e_t - e_{t-1}$$

Nr = summation of difference of errors squares

$$= d_1^2 + d_2^2 + d_3^2 + \dots + d_t^2 =$$

$$(e_2 - e_1)^2 + (e_3 - e_2)^2 + \dots + (e_t - e_{t-1})^2$$

Dr = sum of square of errors

$$DW = \frac{\sum_{t=2}^n (e_t - e_{t-1})^2}{\sum_{t=1}^n e_t^2} = \frac{(\text{sec}_{\text{error}} - \text{first}_{\text{error}})^2}{SSE}$$

x	y	$y_p = -2.6268x + 1,129.2$	error
10	1100		
20	1200		
35	985	Type equation here.	
40	750		
50	1215		
45	1000		

x	y	$y_p = -2.6268x + 1,129.2$	$\text{error} = (y_a - y_p)$
10	1100	$-2.6268 * 10 + 1,129.2 = 1102.9$	-2.9
20	1200	$-2.6268 * 20 + 1,129.2 = 1076.7$	123.3
35	985	$-2.6268 * 35 + 1,129.2 = 1037.3$	-52.3
40	750	$-2.6268 * 40 + 1,129.2 = 1024.1$	-274.1
50	1215	$-2.6268 * 50 + 1,129.2 = 997.8$	217.1
45	1000	$-2.6268 * 45 + 1,129.2 = 1011$	-11

DR = sum of square errors

$$e_1^2 + e_2^2 + e_3^2 + e_4^2 + e_5^2 + e_6^2$$

$$(-2.9)^2 + (123.3)^2 + (-52.3)^2 + (-274.1)^2 + (217.1)^2 + (-11)^2 = 140,330.81$$

Nr = Difference of errors

$$d_1^2 = (e_2 - e_1)^2 = (126.2)^2$$

$$d_2^2 = (e_3 - e_2)^2 = (-175.6)^2$$

$$d_3^2 = (e_4 - e_3)^2 = (-221.9)^2$$

$$d_4^2 = (e_5 - e_4)^2 = (491.3)^2$$

$$d_5^2 = (e_6 - e_5)^2 = (-228.1)^2$$

$$(126.2)^2 + (-175.6)^2 + (-221.9)^2 + (491.3)^2 + (-228.1)^2 = 389,406.71$$

$$DW = \frac{\sum_{t=2}^n (e_t - e_{t-1})^2}{\sum_{t=1}^n e_t^2} = \frac{389,406.71}{140,330.81} = 2.77$$

Dw test interpretation:

Ranges 0 to 4

DW = 0 to 2 : Positive relation

DW = 2 : No relation

DW = 2 to 4: Negative relation

Homoscedasticity :

- Errors have constant variance throughout the data
- Constant variance means homoscedasticity
- The distribution of errors is constant

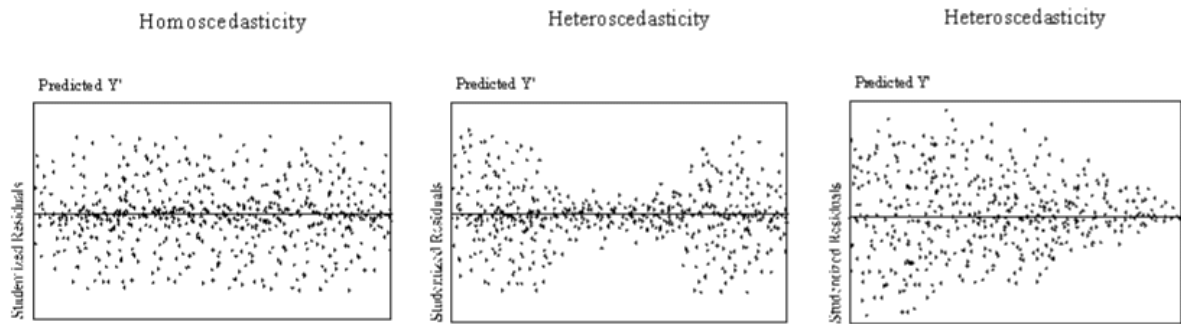
Heteroscedasticity :

- Errors have not constant variance (tunnel shape) throughout the data
- No Constant variance means heteroscedasticity
- The distribution of errors is not constant

How to check:

Residuals vs Fitted plot

Errors vs Predictions



Normality:

Errors should follow Normal distribution

How to check:

Q – Q plot : Quantile – Quantile plot

Assignment : Learn about QQplot



All together

1) *Linearity : we want Linearity between input and output*

Scatter plot

2) *No Multicollinearity : we dont want multicollinearity between input columns*

VIF

3) *No autocorrelation: We dont want relation between errors*

Dw test

4) *Homoscedasticity: we want constant variance of errors*

Residual vs fitted value or Errors vs Prediction

5) *Normality: we want error follows Normal distribution*

QQplot